StockFlow

Inventory management with a stock movement ledger

Piyush

DevSecOps capstone | 16 October 2026

Deployed and verified on AWS EKS in Sydney.

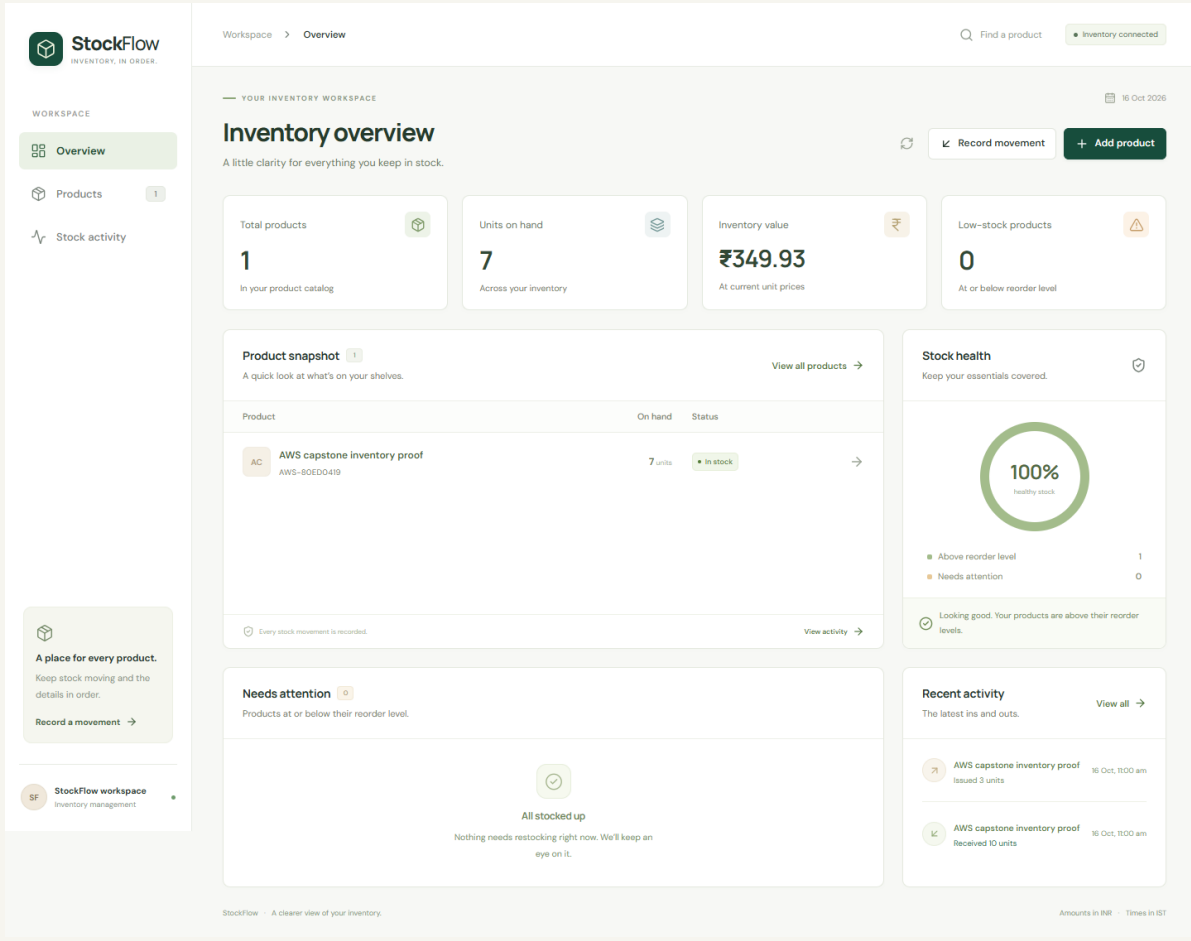
Inventory workflow

Product catalog

Receipts and issues

Append-only movement history

Low-stock reporting



Source: docs/CLOUD_RUNTIME_VERIFICATION.json | Actual public AWS application

Application architecture

React / Nginx → **FastAPI** → **PostgreSQL 16**

Browser calls use relative /api paths.

Alembic applies the schema before API readiness.

Row locks serialize competing stock changes.

Inventory and ledger writes share one transaction.

Source: backend/app/, backend/alembic/, docker-compose.yml

Tests and local containers

49 backend tests | 11 frontend unit tests | 8 browser workflows

Real PostgreSQL tests cover concurrency and rollback.

Disposable test databases protect development inventory.

Compose starts the database, migration, API, and UI.

Both application images run as non-root users.

Source: docs/CI_RELEASE.json, docs/FRONTEND_VERIFICATION.md, docs/CONTAINERS.md

CI and image security

Tests > image build > Trivy gate > SHA-tagged GHCR images

HIGH/CRITICAL image findings fail publication.

The pipeline publishes the same scanned image bytes.

Pull requests receive no package publication credentials.

Terraform has three separate documented scan findings.

Source: .github/workflows/ci.yml and docs/CI_SECURITY.md

AWS infrastructure

Sydney | EKS | private worker | NAT | public ALB | gp3

EKS ACTIVE; five active managed add-ons.

One private m7i-flex.large; encrypted gp3 storage.

Scoped Pod Identity roles; no worker redundancy.

Full demo estimate: $0.33/hour plus variable usage.

Source: docs/AWS_VERIFICATION.json and docs/DEPLOYMENT_REVIEW_20261016.json

Kubernetes and Helm

Two Ready frontend and backend replicas each.

Encrypted Bound gp3 PVC and ordered migration Job.

Helm upgrade and database replacement retained stock.

Public ALB frontend and API targets are healthy.

Source: docs/CLOUD_HELM_VERIFICATION.json and docs/AWS_VERIFICATION.json

GitOps release and rollback

Green source CI > release-values commit > ArgoCD sync

Initial sync: Synced and Healthy, with migration hooks.

A Git promotion changed both published image SHA tags.

Reverting that commit restored the previous image release.

Quantity 7 and the same PVC survived both changes.

Live monitoring



4,601 read requests; 0 failures; HPA observed 2 > 3 ready replicas

Source: docs/CLOUD_OBSERVABILITY_VERIFICATION.json | Actual EKS Grafana

Recovery and lessons

Backend and database replacement Pods became Ready.

Quantity 7 survived release, rollback, and recovery.

ArgoCD sync must preserve database migration hooks.

Managed project policy required EKS Pod Identity.

Source: docs/CLOUD_RUNTIME_VERIFICATION.json and docs/CLOUD_VERIFICATION.md

Evaluation and cleanup

Silent walkthrough explains the project with captions.

Keep the live demo for the agreed evaluation window.

AWS continues using credits: about $8/day plus usage.

Back up data, remove the ALB/workloads, then destroy Terraform.

Open live StockFlow | Destroy removes this public URL.

Source: docs/CLOUD_VERIFICATION.md and demo/RECORDING.md